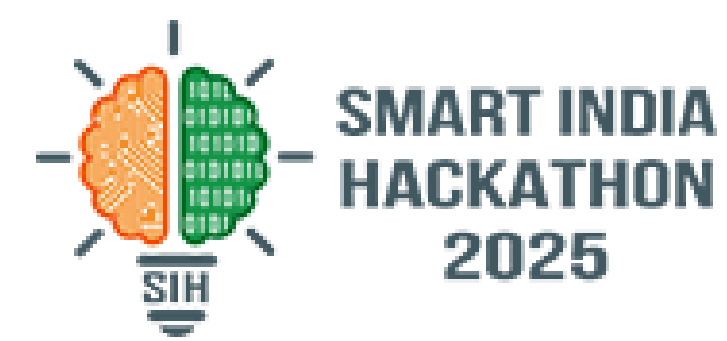
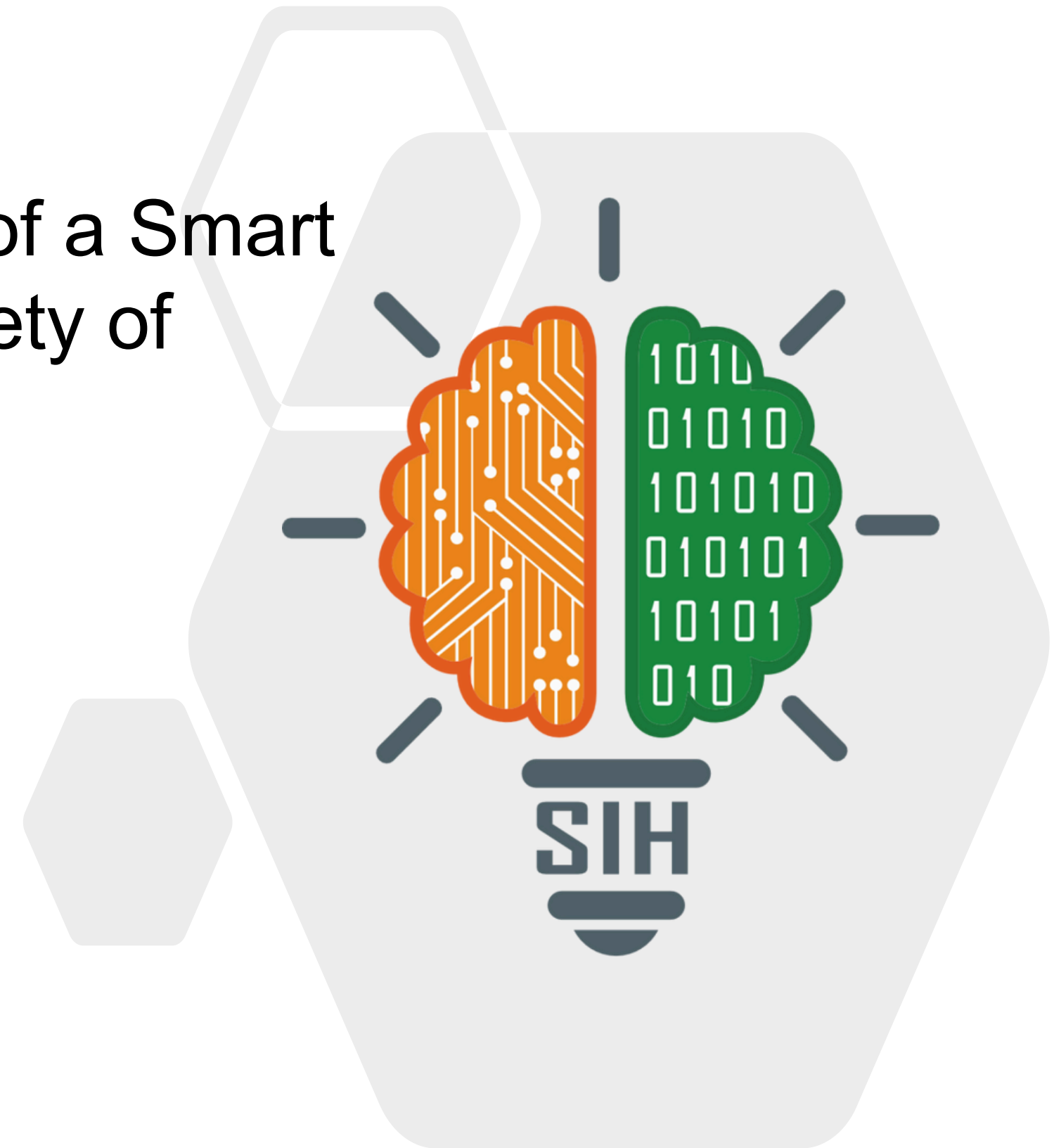


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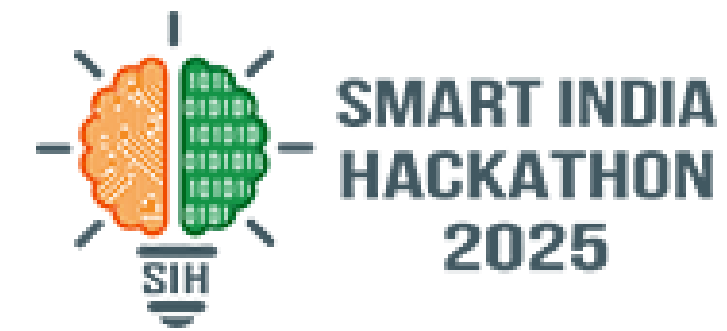


- **Problem Statement ID – SIH25145**
- **Problem Statement Title-** Development of a Smart Blind Stick for Enhanced Mobility and Safety of Visually Impaired Persons
- **Theme-** MedTech / BioTech / HealthTech
- **PS Category-** Hardware
- **Team ID-** 79242
- **Team Name-** TEAM_CHECKMATES





Smart Blind Stick



Problem Statement

- Visually impaired individuals struggle with safe navigation due to the limitations of traditional white canes.
- These canes cannot detect head-level obstacles, provide no real-time alerts, and do not support emergency communication.
- There is a need to design an affordable Smart Blind Stick that enhances navigation, safety, and independence using modern technology.

Proposed Solution

GUIDAR:
Your Guide to Independence



Intelligent mobility that sees, thinks, guides, and empowers true independence

Innovation



From smart cane to autonomous guide rover - detection evolved into physical steering



Smart Blind Stick



Survey at: National Blind Youth Association

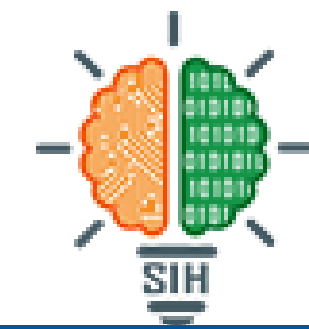
We conducted a focused survey with members of the National Blind Youth Association (NBYA) to understand mobility challenges faced by visually impaired youth.

Short Documentary of Our Survey



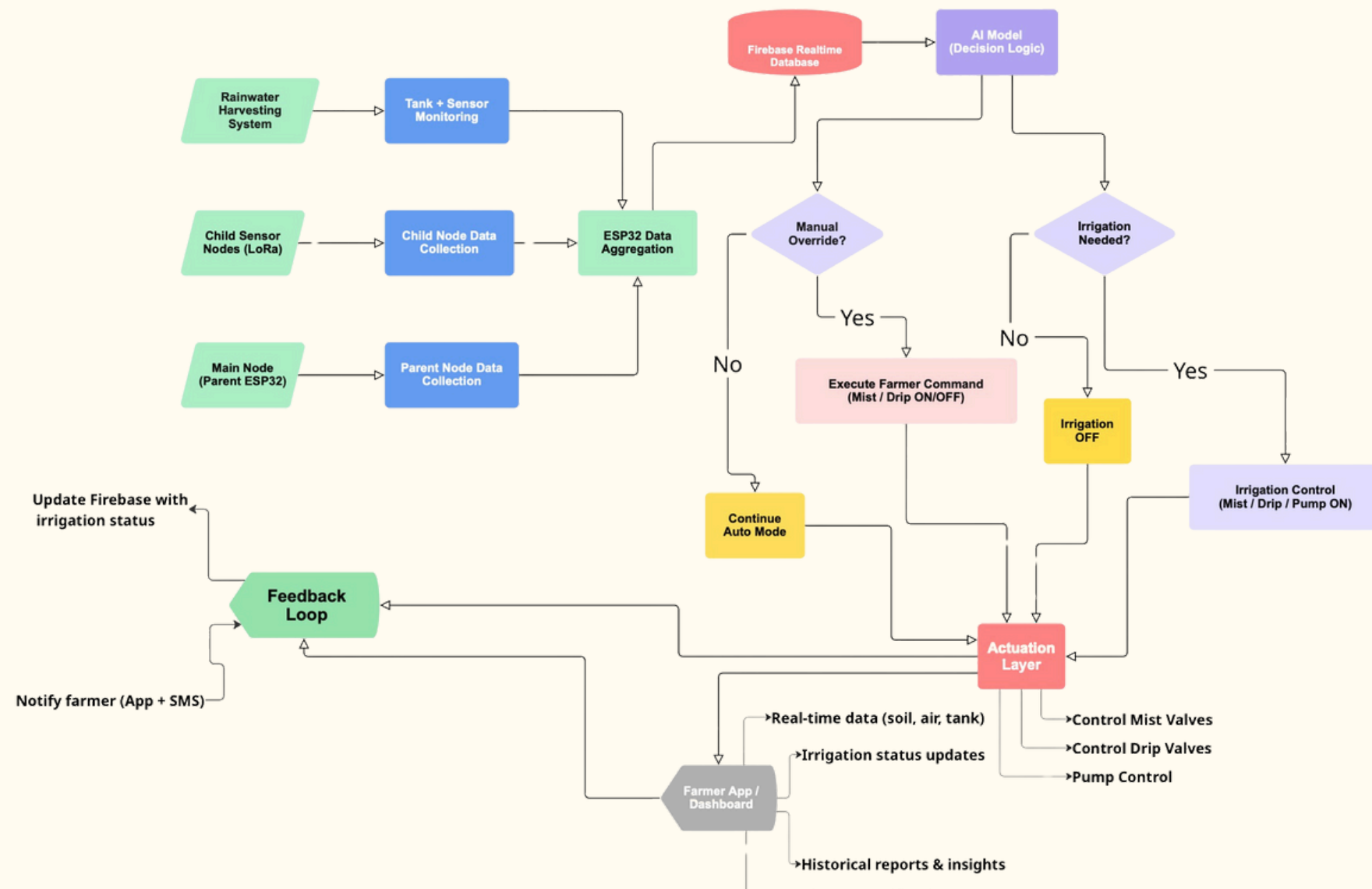


TECHNICAL APPROACH



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Project flowchart



Techologies Used

SOFTWARE

Mobile App:

- Flutter - Cross-platform framework (Android & iOS)
- Node.js - Backend server
- Firebase - Real-time database, messaging, authentication

AI/ML:

- LAMMA LLM - Language model
- Moondream model
- YOLO model

HARDWARE

Main Processing:

- Atmega microcontroller
- ESP32 - WIFI/Bluetooth connectivity
- Raspberry Pi 4 - 8GB RAM (Main computer)

Programming:

- C/C++ - Embedded programming
- Arduino IDE - Development environment

Sensors:

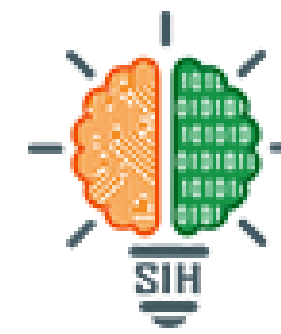
- ToF (Time of Flight) sensors - Distance measurement
- Ultrasonic sensors - Obstacle detection

Additional Components:

- USB Camera - Computer vision
- PCB - Custom printed circuit board
- Motors - Movement and steering

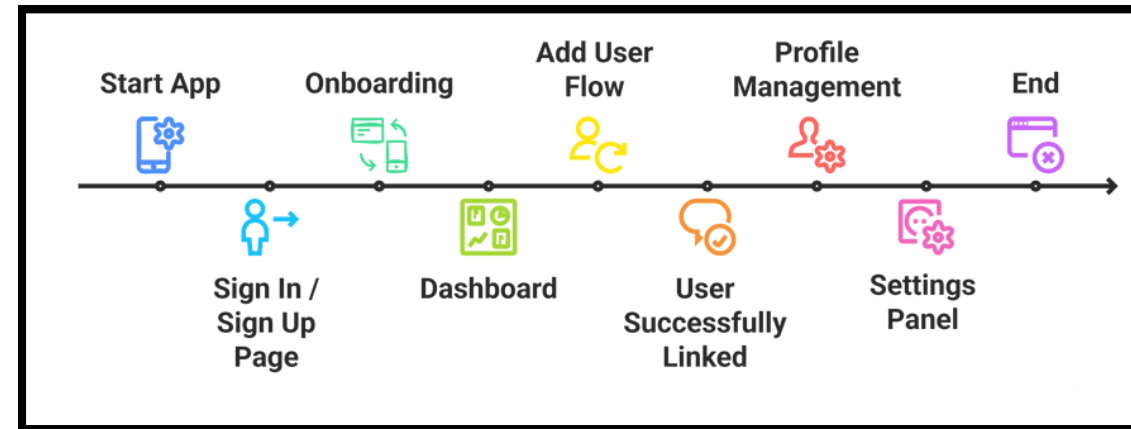


Smart Blind Stick Application

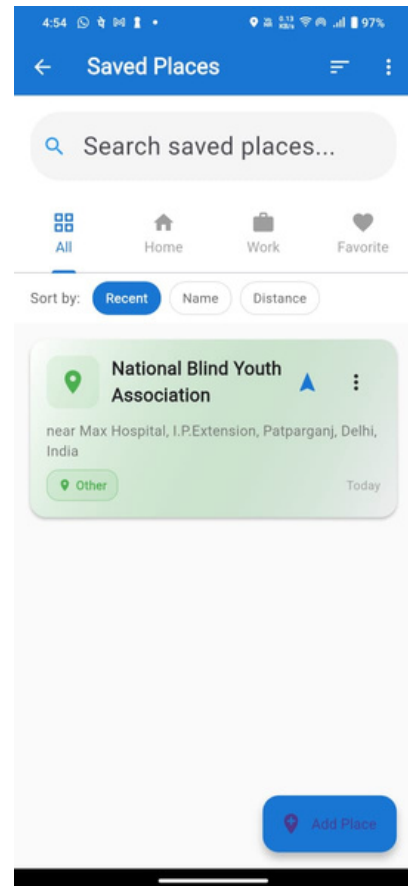
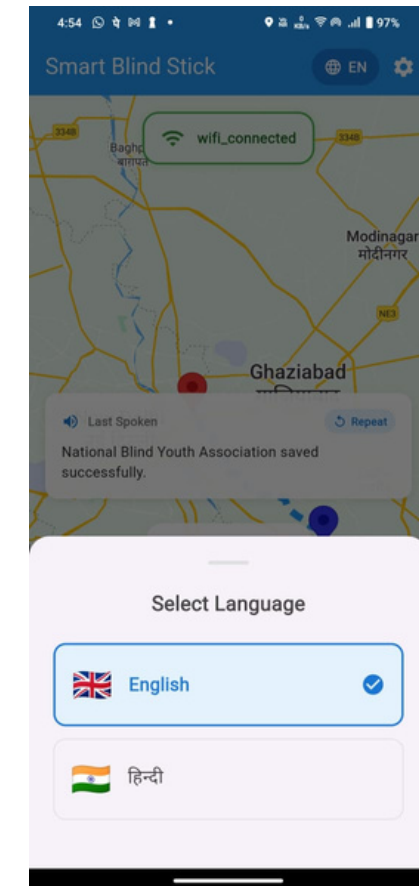
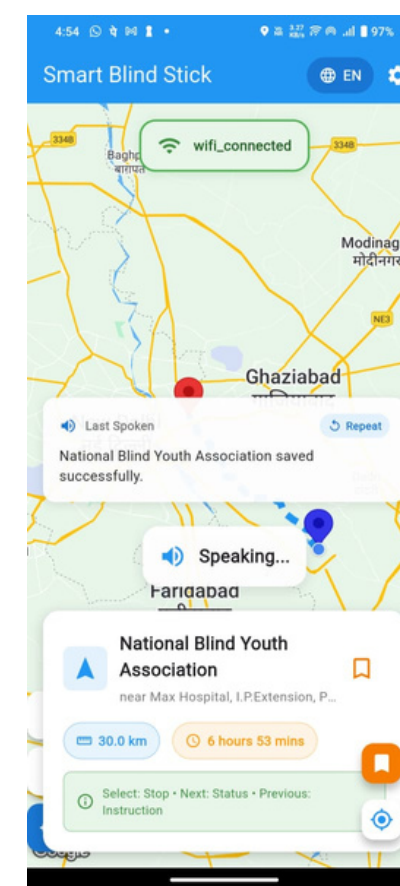
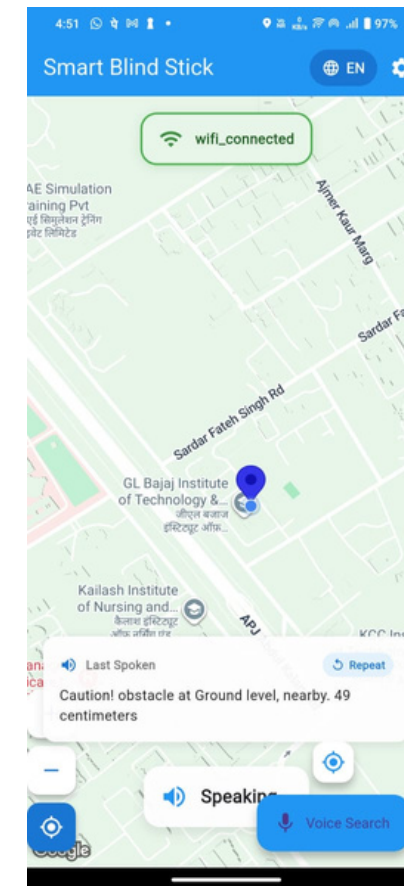
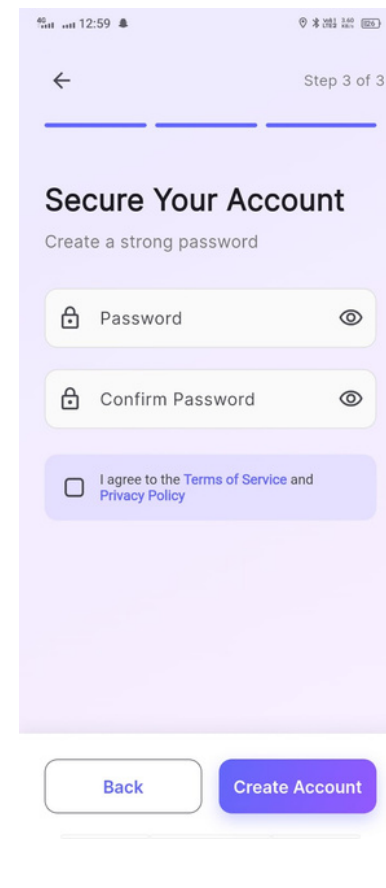
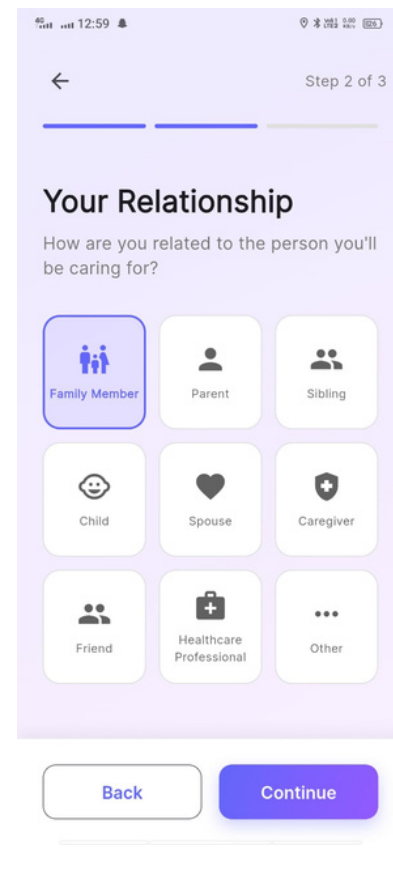
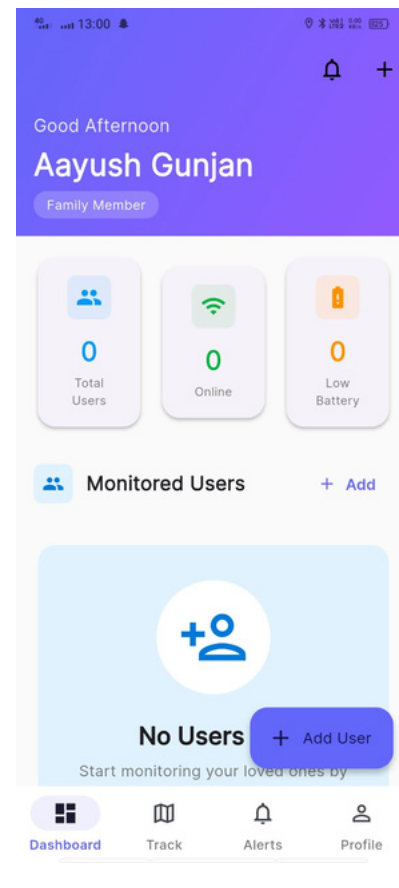
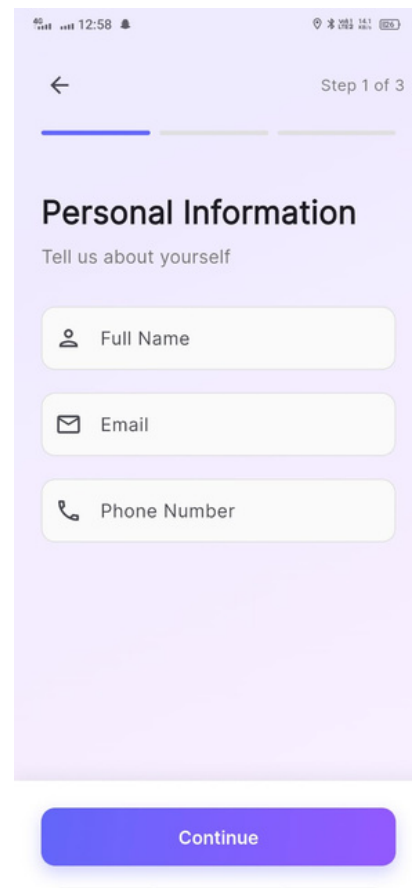
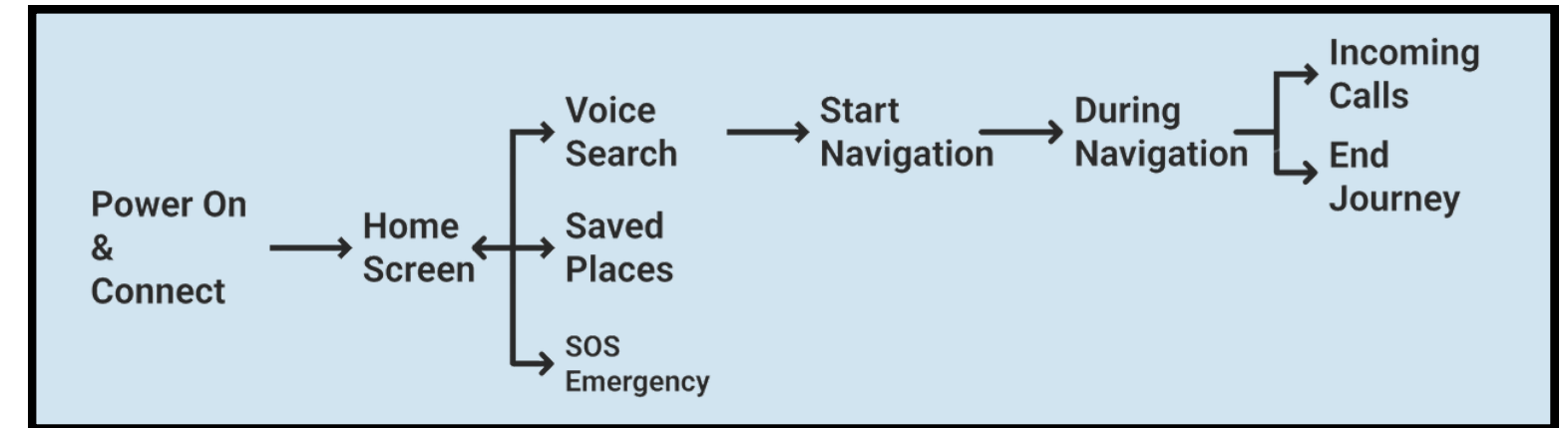


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App flow for Caretaker



App flow for blind



How we stand out?

GUIDAR

Full-Body, Head-Level Hazard Intelligence

AI-Driven Navigation & Smart Services

Dual-App Ecosystem for User & Care Taker

Rover Stability With Ergonomic Control

Unmatched Independence & Reliability

Our USP

Edge AI Vision + Multi-Sensor Awareness

On-device AI with Camera, TOF, Ultrasonic, and IMU delivering instant 360° environmental awareness.

Dual-Mode Navigation (Indoor + Outdoor)

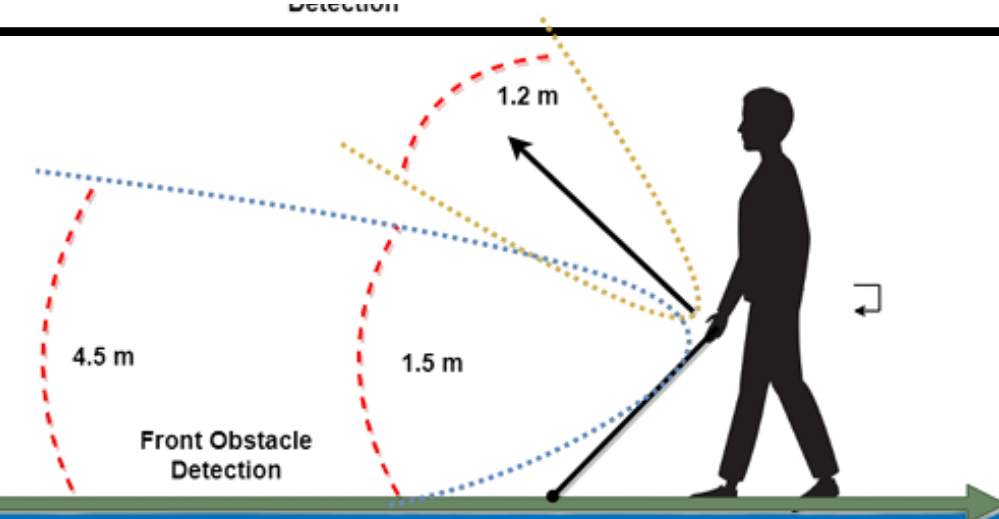
Matrix-logic indoor mapping and GPS-GSM outdoor routing for seamless continuous navigation.

Multimodal Guidance + Smart Services

Matrix-logic indoor mapping and GPS-GSM outdoor routing for seamless continuous navigation.

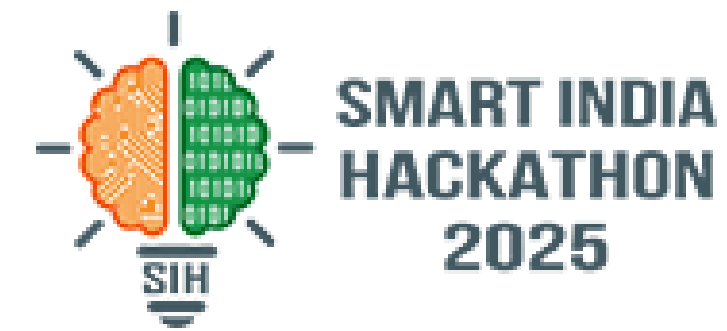
Intelligent Rover Design & Ecosystem

AI audio and multidimensional haptic feedback enhanced with AI navigation, and SOS connectivity.





Smart Blind Stick



GUIDAR:

- **Guide Stick**- Physical Steering
- **360° Multi-Sensor Detection**
- **GPS Navigation with Voice Guidance**
- **Mobile App with Real-Time Monitoring**
- **Emergency SOS with Location Sharing**
- **India-Optimized (Potholes, Hindi, Crowds)**



Phoenix Medical:

No GPS, no app, no SOS, traditional stick only



Saarthi:

No navigation, no app, no emergency system, basic detection



WeWalk:

Traditional stick (not guide robot), no 360° detection, no ai features

Challenges

- Maintaining stable Wi-Fi and GPS outdoors.
- Achieving accurate voice recognition in noisy areas.
- Detecting obstacles reliably across different environments.
- Managing battery life for long usage.
- Ensuring SOS works even in low-network zones.

Feasibility

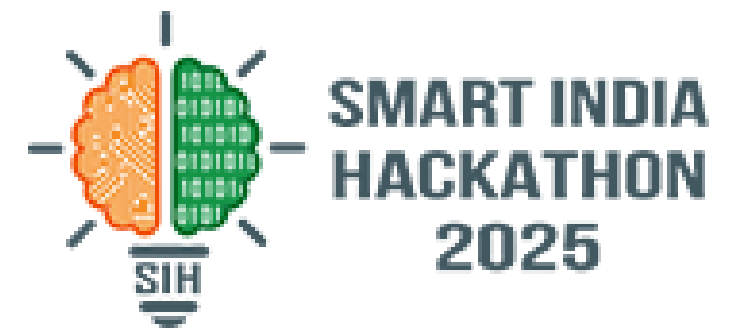
- Uses easily available hardware (ESP32, sensors, vibration motor)
- App built on Flutter makes cross-platform development simple.
- Wi-Fi hotspot connectivity removes need for complex networking.
- Features rely on stable, existing APIs (Maps, TTS, STT, SMS).
- Prototype testing shows reliable navigation and SOS performance.

Viability

- Solves a real mobility challenge for visually impaired users.
- Low-cost hardware makes the solution widely accessible.
- SOS, navigation, and call features add strong daily-life value.
- Scalable for future features (health, indoor navigation, more languages).
- High potential for adoption by NGOs, schools, and government programs.



Smart Blind Stick Snapshots



Final Product

